Creatinine

Test Details

METHODOLOGY

Kinetic Jaffe

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Test Includes

Creatinine, serum; eGFR calculation

Use

A renal function test used in eGFR calculation.[2]

High creatinine: Renal diseases and insufficiency with decreased glomerular filtration, urinary tract obstruction, reduced renal blood flow including congestive heart failure, shock, and dehydration; rhabdomyolysis can cause elevated serum creatinine. ***Low creatinine:*** Small stature, debilitation, decreased muscle mass; some complex cases of severe hepatic disease can cause low serum creatinine levels. In advanced liver disease, low creatinine may result from decreased hepatic production of creatinine and inadequate dietary protein as well as reduced muscle mass.[3]

Limitations

With reduced renal blood flow, creatinine rises less quickly than urea nitrogen. Concentration of creatinine only becomes abnormal when about half or more of the nephrons have stopped functioning in chronic progressive renal disease. Antibiotics containing cephalosporin lead to significant false-positive values if samples are drawn within four hours of a dose.[4] With severe renal disease, creatinine is not reliable in the presence of cefoxitin therapy. There is less interference reported from the cephalosporins cephalothin, cephaloridine, cephadrine sodium, and cephaloglycin dihydrate. Lipemia, hemolysis, and bilirubin may interfere.[5,6]

Interfering Endogenous Substances[7]	**Deviations**	**Degree of Interference**
Pyruvate (serum/plasma)	>2.6 mg/dL	+12.8%
Glucose (serum/plasma)	>450 mg/dL	+16%
Ascorbic acid (serum/plasma)	>88 mg/dL	-10.1%
Urea (urine)	>12612 mg/dL	-11.0%

Specimen Requirements

Specimen

Serum (preferred) **or** plasma

Volume

1 mL

Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube, gel-barrier tube, **or** green-top (lithium heparin) tube. Do **not** use oxalate, EDTA, or citrate plasma.

Collection Instructions

Separate serum or plasma from cells within 45 minutes of collection.

Stability Requirements

Temperature	Period
Room temperature	14 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Reference Range

See table.

Age	Female* (mg/dL)	Male (mg/dL)
Age/sex regardless	0.57−1.00	0.76−1.27
<2 m	0.44−1.19	0.44−1.19
2 to 11 m	0.17−1.18	0.17−1.18
1 to <3 y	0.19−0.42	0.19−0.42
3 to <5 y	0.26−0.51	0.26−0.51
5 to <7 y	0.30−0.59	0.30−0.59
7 to <9 y	0.37−0.62	0.37−0.62
9 to <11 y	0.39−0.70	0.39−0.70
11 to <13 y	0.42−0.75	0.42−0.75
13 to <15 y	0.49−0.90	0.49−0.90
≥15 y	0.57−1.00	0.76−1.27
*Labcorp internal studies.		

Storage Instructions

Maintain specimen at room temperature.[1]

Causes for Rejection

Hemolysis; improper labeling

Footnotes

1. Labcorp internal studies.

2. Levey AS, Stevens LA, Schmid CH; CKD-EPI (Chronic Kidney Disease Epidemiology Collaboration). A new equation to estimate glomerular filtration rate. *Ann Intern Med.* 2009 May 5; 150(9):604-612. PubMed 19414839 (http://www.ncbi.nlm.nih.gov/pubmed/19414839)

3. Takabatake T, Ohta H, Ishida Y, Hara H, Ushiogi Y, Hattori N. Low serum creatinine levels in severe hepatic disease. *Arch Intern Med.* 1988 Jun; 148(6):1313-1315. PubMed 3377614 (http://www.ncbi.nlm.nih.gov/pubmed/3377614)

4. Durham SR, Bignell AH, Wise R. Interference of cefoxitin in the creatinine estimation and its clinical relevance. *J Clin Pathol.* 1979 Nov; 32(11):1148-1151. PubMed 512029 (http://www.ncbi.nlm.nih.gov/pubmed/512029)

5. Bowers LD, Wong ET. Kinetic serum creatinine assays. II: A critical evaluation and review. *Clin Chem.* 1980 Apr; 26(5):555-561. PubMed 7020989 (http://www.ncbi.nlm.nih.gov/pubmed/7020989)

6. Soldin SJ, Henderson L, Hill JG. The effect of bilirubin and ketones on reaction rate methods for the measurement of creatinine. *Clin Biochem.* 1978 Jun; 11(3):82-86. PubMed 688598 (http://www.ncbi.nlm.nih.gov/pubmed/688598)

7. Roche Reagent Bulletin. TP-00523 Creatinine Jaffe Assays-Updated Endogenous Interference Claims. Dec 26, 2018.